Hall Ticket Number:

MR-21

Code:B414CS3S3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Database Management Systems

(Common to CSE and IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 =20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What do you mean by Logical Data Independence?	2	1	1	1,2
1. b)	What is Composite Attribute? Give an example.	2	2	1	1,2
1. c)	Compare Procedural and Non- Procedural Query Languages.	2	4	2	1,2
1. d)	What is the purpose of Join operations?	2	1	2	1,2
1. e)	What is Trivial Functional Dependency?	2	1	3	1,2
1. f)	Why Decomposition is important? Also Explain the types of Decomposition.	2	3	3	1,2,3
1. g)	Define a Schedule? Give an Example.	2	1	4	1,2
1. h)	What is Transaction? Explain with example.	2	2	4	1,2
1. i)	What is Sparse index? Give an example.	2	2	5	1,2
1. j)	Classify the Hash Indexing techniques.	2	2	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain Data Abstraction. Why is it important?	3	2	1	1,2
2. b)	What is ER diagram? Discuss various symbols used in ER diagram. Draw and Explain the ER diagram for Library Management System.	7	3	1	1,2,3
	OR				
2. c)	Explain the Advantages of Database management system over File System.	10	2	1	1,2
3. a)	Explain the Difference between Nested Queries and Correlated Queries. Explain with the help of an example.	5	2	2	1,2,3
3. b)	What is Referential Integrity? Explain its enforcement in DBMS by illustrating with an example SQL statements.	5	3	2	1,2
	OR				
3. c)	Consider the following relations: Hotel (hotelNo, name, address) Room (roomNo, hotelNo, type, price) Booking (hotelNo, guestNo, dateFrom, dateTo, roomNo) Guest (guestNo, name, address) Write the SQL statements for the following: i) List all family rooms with a price below Rs.400, in ascending order of price. ii) Display the number of hotels. iii) Display guestNo, name, address and roomNo details of guests booked room from 01-Jan-23. iv) Display hotelNo and the number of rooms in each hotel.	10	3	2	1,2,3
4. a)	What is Normalization? Differentiate between BCNF and 3 NF with proper example.	5	4	3	1,2,3
4. b)	What do you mean by redundancy? Discuss about insertion, updation and deletion anomalies.	5	2	3	1,2
	OR				

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4. c)	Given R (A, B, C, D, E, F) and the set of FDs: {A → BC, E → CF, B → E, CD → EF}. Compute the CLOSURE $\{A, B\}^+$ under the given set of FD's. Show that an FD: AB → BCE hold for R.	10	3	3	1,2,3
5. a)	Why do we need Concurrency Control? How can you control the concurrent access of database using Locking? Explain the Two-Phase Locking Protocol.	10	2	4	1,2
	OR				
5. c)	Discuss the various States of a Transaction with the help of a neat diagram.	5	2	4	1,2
5. d)	Discuss about Conflict Serializability with an example.	5	2	4	1,2
6. a)	What is Multi-Level Indexing? How does Multi-Level Indexing improve the efficiency of searching an index file?	5	2	5	1,2
6. b)	Distinguish between Primary Index and Secondary Index. Illustrate with an example.	5	4	5	1,2
	OR				
6. c)	List and discuss about various File Organization Techniques with an example.	10	2	5	1,2,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome

Hall Ticket Number:

MR-21

Code: B414CB3S3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Introduction to Database Management Systems

(Common to CSB, CSM and CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 70

Part-A (10 x 2 =20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Differentiate between DBMS and RDBMS.	2	3	1	1,2
1. b)	What is a weak entity set? How do you convert it into a relation?	2	2	1	1,2
1. c)	What is a super key and a primary key. Write their difference.	2	3	2	1,2
1. d)	How do you create and modify the database?	2	1	2	1,2
1. e)	Define Functional dependency and Multivalued dependency.	2	1	3	1,2
1. f)	When do you say that a relation is in 2NF?	2	1	3	1,2
1. g)	Mention different transaction states.	2	1	4	1,2
1. h)	What is Atomicity and Isolation property of transactions.	2	1	4	1,2
1. i)	How is data organized in a hash based index?	2	1	5	1,2
1. j)	Explain Failure with loss of non volatile storage.	2	1	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the Architecture of DBMS with a neat sketch.	5	1	1	1,2
2. b)	Write about different types of database users.	5	2	1	1,2
	OR				
2. c)	Draw an ER diagram for a banking enterprise and convert it into a set of relations.	5	3	1	1,2
2. d)	Explain the additional features of ER model.	5	3	1	1,2
3. a)	Write about SQL data types and different types of constraints that can be applied on data.	5	3	2	1,2
3. b)	Explain the following Operators in SQL with examples: i) SOME ii) IN iii) EXCEPT iv) UNION v) ALTER	5	1	2	1,2
	OR				
3. c)	Explain DDL and DML commands with examples.	5	1	2	1,2
3. d)	Write the aggregate operators in SQL.	5	1	2	1,2
4. a)	Compute the closure of the following set of functional dependencies for a relation scheme. R(A,B,C,D,E,F,G,H), F={AB->C, BD->EF, AD->G, A->H} List the candidate keys of R.	5	3	3	1,2
4. b)	Explain 3NF and BCNF with examples. What is the difference between them?	5	3	3	1,2
	OR				
4. c)	Write the need for schema refinement in relational database design.	5	4	3	1,2
4. d)	Define MVD? Explain 4NF with an example.	5	2	3	1,2
5. a)	Explain Time stamp based protocol.	5	2	4	1,2
5. b)	Explain Log based recovery	5	2	4	1,2
	OR				
5. c)	Explain Transaction concept.	5	2	4	1,2
5. d)	What is ARIES algorithm? Explain.	5	2	4	1,2

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6. a)	Explain various file organization methods.	5	2	5	1,2
6. b)	Compare static hashing and Extendible hashing.	5	4	5	1,2
	OR				
6. c)	Construct a B+ tree to insert the following key elements (order of the tree is 3) 6, 11, 12, 19, 21, 25, 32, 36, 38, 42, 51, 55, 67, 70.	5	3	5	1,2
6. d)	Explain cluster Index.	5	2	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome